

Rules vs. model — the actual measured result

The comparison `docs/MODEL_BATTERY_PLAN.md` §2 describes, run for real on 2026-09-18: `openai/gpt-4o-mini`, live, full 50-case / 118-trial eval set, run twice — once with `DECISION_MODE="rules"`, once with `"model"` — nothing else different. This is the actual evidence behind the team's D0/D6 architecture argument, not a projection.

Raw results committed at `A2_scaffold/results_rules_gpt4o-mini.json` and `A2_scaffold/results_model_gpt4o-mini.json`. Reproduce the table below with `python3 A2_scaffold/compare_modes.py results_rules_gpt4o-mini.json results_model_gpt4o-mini.json`.

The headline numbers

	rules	model
trials	118	118
pass rate (exact code check)	78.0%	37.3%
cost (full run)	US\$0.1084	US\$0.1311
median turns	2.0	2.0
worst-case turns	4	5
<code>route_mismatch</code> (blocked a book)	0	0
<code>route_mismatch_nonblocking</code> (logged only)	12	17

Nothing hit a guardrail in either run (`stopped_by` was `null` for all 236 trials) - every difference here is decision quality and cost, not loop failures or budget/step caps.

Do not stop at the headline number - it is misleading on its own

Broke down all 74 failing `model`-mode trials by what actually mismatched:

Category	Count	What it means
Decision itself was wrong	26	A real reasoning miss - the agent concluded the wrong outcome.
Decision right, trigger <i>label</i> wrong	47	e.g. <code>'sudden visual loss'</code> instead of the answer key's <code>'red_flag_term'</code> - same finding, different words.
Decision right, booked slot wrong	1	Right call to book; wrong clinic/date/time.

If you set aside the trigger-labelling mismatches (where the decision itself was correct), `model` mode's real decision-level accuracy is $44 + 47 + 1 = 92/118 = 78.0\%$ - statistically identical to `rules` mode.

Why the gap is mostly a labelling artifact, not a reasoning gap

"rules" mode hands the model `resolve_routing()`'s already-computed `trigger` string and tells it to report that value faithfully - `prompt.py`'s `RULES["B_rules"]` text is explicit about this. "model" mode gives the model only the raw facts (`red_flag_term`, `right_department`, `missing_tests`, `existing_appointments`) and asks it to derive and *name* the trigger itself, in its own words. The answer key's `code_check()` does an exact string match on `trigger`. A model that correctly identifies "there's a red flag, escalate" but writes `'sudden visual loss'` (the actual red-flag phrase it saw) instead of the category name `'red_flag_term'` gets marked wrong by the code check even though a human reading the record would call it right - which is exactly what the `must_record` judgement-check items are for (see `docs/HOW_IT_WORKS.pdf`'s section on the two kinds of check).

What this actually supports for the D0/D6 argument

- **Rules mode's real, defensible advantage is reliability of exact output and lower cost** - not better underlying reasoning. `gpt-4o-mini` reasons to essentially the same decision either way; it just doesn't know this assignment's specific trigger vocabulary unless it's told.
- **The 26 genuine reasoning misses (22% of all trials) are still real** and worth a closer look in the report - these are `model` mode getting the actual `OUTCOME` wrong, not just the label.
- **Cost favours rules mode too** (\$0.1084 vs \$0.1311) - a shorter, more directive prompt plus not needing to re-derive four gates from scratch.
- `route_mismatch_nonblocking` (**17 vs 12**) confirms "model" mode disagrees with the mechanical protocol-only answer more often than "rules" mode does - expected, since "rules" mode is largely just reporting that same mechanical answer back.

Caveats - keep these attached to the number

- **One cheap-tier model only.** This is `gpt-4o-mini`'s specific trigger-labelling behaviour, not a universal law about rules vs. model reasoning. A different model (especially a more capable or instruction-tuned one) may label triggers more consistently with the answer key, closing some or all of this gap.
- **Not the cross-model battery.** This is the separate, single-model rules-vs-model comparison from `docs/MODEL_BATTERY_PLAN.md` §2 - do not present it as D5(b) cross-model evidence.
- **Report BOTH numbers.** 37.3% (strict code check) and 78.0% (decision-only, trigger-wording set aside) are both real and both belong in the write-up - reporting only one tells a misleading story in either direction.

Written up 2026-09-18. See `docs/CHANGELOG.md` for the JSON-compliance fix and the unknown-tool-name fix that made this run possible - the first two attempts at this exact comparison were

unusable data, not real measurements. See docs/MODEL_BATTERY_PLAN.md for the frontier-tier (claude-opus-5) result, run separately.